

Assignment 4B: Research Paper

Creating Metadata for Children's Resources: Metadata for Children by Children

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LIS 4010: Organization of Information

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Abstract

Children in preschool through the fourth-grade search for books in unique ways that are not always suited to existing metadata schemata. Traditional metadata and organization systems serve librarians, parents, and other adults in assisting younger users because the elements and taxonomies those systems utilize are developmentally appropriate in the adult age range. Adults can comprehend the utility of the divisions whereas children have yet to group the world into academic disciplines.

Researchers suggest addressing this dichotomy's root cause through the creation of special metadata for children using design partnerships where children co-create metadata ensuring its future efficacy. If metadata is meant to support access to resources for the audience a library serves, then there is more that should be done to aid children in independent searching, to encourage the exploratory nature of a child's search strategy, and to support children in finding materials of interest in children's libraries.

Part I Overview – It's Not Your Data, It's Just Metadata

When organizing any collection of library materials, it is generally accepted that one of the most important factors to consider is the needs and information seeking behaviors of the users of that collection. However, organizational systems and their associated metadata have historically been created in the context of adult access. In simple terms, metadata and classification structures have been created and organized by adults for adults, even when the primary audience of the collection is children. While this makes sense for the parents, guardians, volunteers, and librarians attempting to assist younger children in their information retrieval activities, researchers question whether these structures directly serve children themselves in physical and digital search settings.

In their journey toward independent information retrieval, the prevailing thought used to be that if young children are introduced to Dewey Decimal Classification (DDC) early on, they will be at an advantage for finding needed materials in larger libraries later in their educational careers. However, contemporary thinking points to three problems with DDC. First, children do not yet think in terms of the academic disciplines DDC is organized into. Second, DDC holds biases that are largely outdated and primarily derived from Western viewpoints and is therefore not suitable for all children. Finally, DDC is a numerical code that children are not drawn in by (Kaplan et al., 2013). Furthermore, in terms of educational levels, children in the United States do not learn the decimal system needed to grasp DDC until the end of the fourth grade (Almond, 2023). Consequently, in combing through educational blogs, we see that many elementary school librarians feel justified in not teaching DDC. In a pilot study on children's book selection, Beak (2012) noted that current knowledge organization systems such as DDC "are created in a resource-centered view rather than in consideration of children's cognition" (p. 1). This means that the organization system is designed to convey metadata about information resources using elements such as author, title, and publication date. It is not designed to describe the elements a young

child may be interested in while browsing or from recall of a previous book they've read such as main character animals, interesting items, cover color, or aesthetically pleasing illustrations. Therefore, this paper is best understood in the context of metadata creation, information retrieval, and classification schemata for children in preschool through the fourth grade. Insights are also provided as to how academicians and practitioners have demonstrated that stand-alone organizational systems such as DDC don't serve the needs of elementary aged children.

Part II Observations – I *HAVE* Metadata I Did Not Like

Evidence suggests that helping children in this lower elementary age range to be independent seekers of knowledge, to enjoy using the library, and to be strong readers requires that concerns about access hindrances, low circulation, and book choice behaviors continue being addressed and explored. For instance:

Children have been found to show greater interest in reading when they are allowed to select their own books rather than receiving teacher-assigned basal readers. Many educators share the experience that children are often able to read beyond their supposed reading levels when they choose reading material for themselves... Previous research suggests that simply by offering children opportunities to choose their own books without any constraints, librarians, educators, and parents may provide greater motivation for children to read (Reuter, 2007, p. 1757).

With regards to classification systems, many school librarians have explored options other than or in combination with DDC that enhance children's ability to locate books of interest without assistance. The classification system used in bookstores, the Book Industry Standards and Communications system (BISAC) is sometimes used alongside DDC, the efficacy of which has mixed reviews. For example, Kaplan et al. (2013) remarked that BISAC is an improvement but is also not ideal because of its non-linear

arrangement. In other words, In BISAC, unrelated topics such as “adventure”, “animals”, and “art” are found next to each other.

Another organization system called Metis was designed with children in mind and uses “whole language, visual cues, and logic that reflects the experiences of a child” to facilitate independent book selection. It was developed by the Ethical Culture School in New York in 2011 (METIS, n.d., para. 3). Metis does a lot to improve the search experience for younger children using picture labels for users who are not yet reading, a “dots and stars” sticker system to denote fiction and non-fiction books that are shelved together according to their primary category, and a Kindergarten through fourth grade centered categorization schedule formed with children’s search behavior in mind (Kaplan et al., 2013).

While combination and alternative or “creative” cataloging like BISAC/DDC and Metis serve to strengthen the nexus between children’s search behaviors and the ability to find an item of interest, they do not address the root issue of creating metadata that caters to user perspectives. To clarify, metadata schema creation differs from metadata record creation and “few age-appropriate controlled vocabularies exist for use in creating metadata” (Abbas, 2005, p. 303). To fully allow children to find books independently and “without any constraints,” a metadata schema designed predominantly for the allowance of self-led book selection and containing elements that children naturally use while searching must be considered. It should be admitted though that creating a universal controlled vocabulary for children’s literature would be a considerable undertaking. One challenge in achieving this goal was discussed by Hurley et al. (2023) who explain that “when we are involved, unknowingly, in creating and perpetuating a problem, we can have a strong motivation to think that it isn’t really a problem. It is learning to see *those* problems that can be transformative” (p.4, emphasis in original). “Those” problems are perpetuated by the fact that librarians know that much of the metadata children utilize to locate books they are interested in is contained in the ‘Notes’ sections of current metadata records. The ‘Notes’

sections are unstructured data. Within unstructured data, keyword searching is required to locate materials of interest. Children sometimes provide keywords to parents and librarians, but this workaround seems to operate as a crutch we use to bypass the systemic issue of metadata not being adequate to describe a child's perspective.

One subset of the absence of metadata for children's literature applies specifically to picture books. Various studies and research articles have helped us see the inherent problems in continuing to utilize metadata elements and controlled vocabulary that are not designed to address children's picture book search behaviors. Reuter and Druin (2004) commented that "by better understanding children's searching and selection behavior, librarians, educators, and information systems designers can design services, instruction, and products to facilitate self-selection, more easily bringing together children and books and encouraging the practice of lifelong reading" (p. 347). The acknowledgement of the need for new elements, vocabulary, and systems indicates a consensus that naturally makes one wonder why existing children's metadata schemata are either not well supported or have not yet gained enough traction to be widely utilized. While part of the answer is certainly systemic, another part of the answer may be found in the fact that modern readers do much of their reading in the fast paced and ever-changing landscape of the online (web) environment.

Another subset of this metadata deficiency is found in digital environments. Hurley et al. (2023) stated that "attempting to redress structural inequities is challenging, and you may face criticism both from those for whom the structures work well, and those who demand quicker and more comprehensive solutions. But if we, as library professionals, can improve our services to everyone in our communities, the work is worth it" (p. 6). In the last decade, a whirlwind of children's online reading applications became available. During the Covid-19 pandemic, digital reading and other educational applications supporting online learning experienced accelerated growth. In 2016, the anticipated growth

rate for these applications was 6% (Technavio, 2016), whereas in 2020, the projected growth rate was 26% (Technavio, 2020). This stark increase illustrates the demand for access to digital children's materials. While these private, for-profit resources grant additional broad access to children whose guardians can afford such services, they typically do not cater to younger children's search strategies, other than browsing. They also do not help to establish a future for increased accessibility via self-selection using a child-centric metadata schema since they typically utilize publisher manufactured metadata.

Whether the application is for profit or not, the web is a more natural landscape within which to address metadata challenges as digital tools proliferate. Even though the observations of all the authors mentioned thus far are born from studies that indicate and/or justify the need for metadata schemas designed with children's search strategies in mind, there are very few operational examples of instances in which such metadata schemas have been deployed successfully and have lasted the test of time. Several of the articles reviewed for this paper mentioned that the International Children's Digital Library (ICDL) employed a metadata schema that allowed children ages 3-13 to search for recreational reading materials in two distinct ways:

The "World" area enabled access to the collection by offering a globe children could "spin" to select books from, about, or set in one of the world's six inhabited continents. The "Categories" area was organized in traditional categories (e.g., subject, genre, age, and publication date) and in novel categories (e.g., characters, color of book cover, and shape) (Reuter, 2007, p. 1748).

Therefore, the ICDL utilized metadata elements and a controlled vocabulary that allowed children to search more holistically for developmentally appropriate markers in areas of their own interest. At the time of this writing, the ICDL no longer functions as described in the excerpt above, though. In April of 2021, the founders posted an explanation as to why many of its original functions can no longer be

supported (International Children's Digital Library, n.d.). It may be of note that this announcement came shortly after the Covid-19 pandemic increased the demand for digital learning solutions.

Part III Options – I Meta New Data

Various examinations of this topic indicate that using metadata elements designed specifically for children using controlled vocabulary gathered from observing children would ease access and encourage reading habits that endure. Thus, intuitive search elements and simple classification systems are hallmarks that could guide future design. A study by Wang et al. (2022) identified three dimensions of picture book components further broken down into 29 metadata elements that outline the types of data children notice and convey to adults in their search efforts (see Appendix A). Reuter (2007) identified seven dimensions of book selection further broken down into 46 factors that outline the types of influence certain elements of books have on children's book choice (see Appendix B). Similarly, Beak (2012) articulates 56 non-exhaustive factors within 14 facets of children's book selection processes (see Appendix C). Beak concludes that "perceptual factors need to be considered when it comes to creating a metadata schema for children's libraries" (p. 9). Whether we call the first level 'dimensions' or 'facets,' and the second level 'elements' or 'factors,' this wealth of child-centric informational clues would ideally be used in the shaping of a controlled vocabulary and the development of metadata schemata that can aid younger children's book selection. It has even been suggested that children be invited in as design partners while writing descriptors in such processes (Martens, 2012).

Digital libraries like the ICDL showed that categories such as color, animal, item, shape, main character, supporting character, location, cover color, and illustration aesthetic, can be turned into structured data and mapped to digital search structures. Thus, a browsing capability with age-appropriate features is another hallmark that should guide future systems design. It should be acknowledged that metadata and systems design for digital libraries and physical libraries may not be

compatible as children's physical search strategies differ from their online search strategies (Beak & Olson, 2011). Complications also arise when considering a universal design that would serve multiple age groups and developmental stages. However, Martens (2012) keenly observed that if redesign is approached thoughtfully and with flexibility in mind, then new systems will not only serve groups of children at their expected ability levels but will also serve users with unique needs as well as marginalized populations.

A final hallmark to consider in future metadata and systems design is an observation that has already been used by teachers in literacy development. The impetus for younger children to respond to a book is less often about the text's utility and more often an internal, experiential process. Many children's authors cater to this knowledge by capturing the imagination of a child first with lovable characters and attractive illustrations while also being sure to incorporate age-appropriate life lessons in a silly or lighthearted manner. For example, Mo Willems' books are well illustrated and humorous, but they also contain messages that help children accept the difficulties of everyday life as a young child. This modus operandi is referred to as "aesthetic reading" wherein a child's interest is internally motivated and serves no particular or immediate goal. Adults are more likely than children to read efferently, meaning that their interest in a book is often directly tied to an external goal, (Reuter, 2007). This was also the impression of Martens (2012) who referred to children's book search behavior as "grazing."

In closing, the research on creating metadata for children's literature encourages stakeholders to pay attention to the unique aspects of children's informational needs and search strategies. In doing so, vocabularies, features, elements, and designs will be able to assist more appropriately. This will, in turn, help children become stronger readers earlier, fall in love with more books, and become lifelong library users. Children may be equally as able as adults to seek information at any age, at any stage of

development, and at any cognition level if the concepts of the metadata and knowledge organization schemes they are faced with have been built for them and with them.

Appendix A

Dimensions and Elements of Young Children's Searches, (Wang et al., 2022)

TABLE 2 Young children's search elements and definitions

Dimension Elements		Definitions	Examples
D1	Animal	An animal	There is a gorilla.
	Motion	A single action performed by a character	It holds a key.
	Item	An item	It holds a key.
	Person	A person	A zookeeper
	Characteristic	The feature of a character or item	Its tail is very long.
	Shape	Shape	It is a rectangle.
	Amount	Amount	There are a lot of cookies.
	Color	Color	That chimpanzee is black.
	Place	A setting	In the zoo
	Chinese character	Chinese characters	I can read, so I know the Chinese characters for "cow" and "milk."
	Layout	The specific layout of a page	Here is a big house.
	Size	Size	That wrong book is too big.
	Time	Time	When they say good night.
	Icon	Icon	A tiger icon.
D2	Main character	A character who is the subject of the story or appears throughout the story	A gorilla and a zookeeper.
	Supporting character	Characters in the story other than the main character	A chimpanzee and its parent
	Key item	An item that can motivate the story or appears throughout the story	The gorilla stole the key.
	Other items	Items in the story other than the key item	The mouse took the slipper.
	Concept	A concept or knowledge in a nonfiction book	A python that can hunt a human
	Scene	The place and time that the story occurs	The gorilla is in a cage.
	Series mark	An icon or pattern representing a picture book from a series of books	They (the correct book and the "error return" book) have the same icon.
D3	Story/subject	The story (fiction)/subject (nonfiction) of a picture book, usually acquired by listening	There is a gorilla, a hyena, and a giraffe.
	Illustration	The illustration of a picture book, usually acquired by vision	The gorilla is in a cage.
	Front cover	Something appears on the front cover.	The cover shows an old woman with white hair, and she is shaking hands with a chimpanzee.
	Title	The title of a picture book	<i>Good Night, Gorilla; If You Feed a Mouse a Cookie</i> (incorrect title)
	Physical characteristics	The physical characteristics of a picture book	That book is big and square in shape.
	Pastedown	Something appears on the pastedown	If you turn the page inside the book, you can see some cookies that look like a mouse had taken a bite of them.
	Title page	Something appears on the title page	There is a crocodile, too, when you open the book.
	Back cover	Something appears on the back cover	There is also a crocodile's tail on the back (of the book).

Appendix B

Dimensions and Factors Influencing Book Selection, (Reuter, 2007)

TABLE 1. Factors influencing book selection, grouped by dimension.

Dimensions and factors	Pairs		Mentions (by factor)		Total mentions (by dimension)	
Metadata & physical entity						
Title	37	77.1%	263	18.0%		
Cover	19	39.6%	38	2.6%		
Front matter	26	54.2%	34	2.3%		
Format	13	27.1%	23	1.6%		
Author/illustrator	7	14.6%	10	0.7%		
Presence of illustrations	7	14.6%	10	0.7%		
Recency	6	12.5%	10	0.7%		
Table of contents	9	18.8%	9	0.6%		
End papers	3	6.3%	3	0.2%		
Contributor	1	2.1%	1	0.1%		
Notes	1	2.1%	1	0.1%	402	27.5%
Accessibility						
Language	38	79.2%	132	9.0%		
Length	37	77.1%	108	7.4%		
Legibility	31	64.6%	51	3.5%		
Quantity	23	47.9%	40	2.7%		
Level of difficulty	13	27.1%	18	1.2%		
Text density	3	6.3%	3	0.2%	352	24.1%
Content						
Illustrations	36	75.0%	105	7.2%		
Summary	21	43.8%	48	3.3%		
Topic	19	39.6%	47	3.2%		
Plot	24	50.0%	33	2.3%		
Genre	14	29.2%	26	1.8%		
Subject headings	4	8.3%	5	0.3%		
Style	4	8.3%	4	0.3%	267	18.3%
Engagement						
Liking	17	35.4%	38	2.6%		
“Coolness”	18	37.5%	31	2.1%		
Mood	16	33.3%	28	1.9%		
Quality	14	29.2%	23	1.6%		
Interest	9	18.8%	12	0.8%		
Reading experience	8	16.7%	9	0.6%		
Visual appeal	5	10.4%	7	0.5%		
Interaction	2	4.2%	3	0.2%	152	10.4%
Novelty						
Curiosity	20	41.7%	52	3.6%		
Language curiosity	21	43.8%	49	3.3%		
Novelty	16	33.3%	28	1.9%		
Unusualness	11	22.9%	14	1.0%		
Variety	7	14.6%	7	0.5%	150	10.3%
Socio-cultural						
Origin	19	39.6%	30	2.1%		
Knowledge sharing	20	41.7%	27	1.8%		
Community ties	11	22.9%	13	0.9%		
Personal connection	6	12.5%	10	0.7%		
Intertextuality	6	12.5%	9	0.6%		
Values	4	8.3%	5	0.3%		
Popularity	3	6.3%	3	0.2%	97	6.6%
Familiarity						
Known item	12	25.0%	25	1.7%		
Previous experience	16	33.3%	18	1.2%	43	2.9%
Totals			1463	100.0%	1463	100.0%

Appendix C

Factors and Facets Related to Children's Book Selection, (Beak, 2012)

Facets	Factors
Contents/Stories	• Topic-Theme • Plot (Actual plot and Imaginary plot by readers) • Narrative style • Setting: When & Where • Table of content • Index • Summary or review from back of books • Specific components or parts of contents
Series	• Titles of series • Characters of series • Subjects or themes of series • Illustrational patterns in series
Characters	• Name of characters • Appearance of characters • Gender of characters • Personality of characters • Occupation of characters
Genres	• Non-fiction • Fiction

	• Chapter books • Picture books • Fairy tales • Fantasy
Illustrations	• Illustrations or particular icons on book covers • Inside illustration • Colors • Aesthetic quality • Characters or objects • Formats of illustrations
Physical characteristics	• Book shape • Text arrangement • Texture on illustrations
Difficulty, Reading level, Understandability	• # of words or pages • Font size • Text size-density • Age appropriateness • Titles • Words inside books
Familiarity	• Prior experiences • Associated media • Recommendation • Language • Book shelves' location in the library
Emotional interest	• Internal factors of person • External factors of books
Personal connection	• Belief or value • Hobbies • Personality
Activities	• Reading programs
Engaging elements	• Hidden letters in illustrations • Folded pages or pictures • Rhythmical narrative styles
Additional materials	• Maps
Basic bibliographical information	• Title • Author • Language

Table 1. Factors and facets related to children's book selection

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Assignment 4A

Creating Metadata for Children's Resources

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Historical Aspects

- 30+ years of research concerning children's access to the catalog
- Research areas:
 - Majority of articles address children's information search behaviors
 - Fewer articles cite accounts of metadata creation that support a child-centric approach
 - Child-centric approaches are either not well supported, used sporadically, lack traction



Current Problems



- Children in the United States learn decimals in the 4th grade. DDC is not compatible with library services for children aged PK – 3rd grade
- Reliance on taxonomies, disciplines, and hierarchical structures limits children's book search strategies
- Publisher provided and copy cataloged metadata records don't contain structured metadata elements that researchers cite as important for children.
 - Children use conceptual elements taught in lower elementary grades such as cover color, main character animal, item that affects the plot (ex: lost baseball), and setting (ex: the zoo)

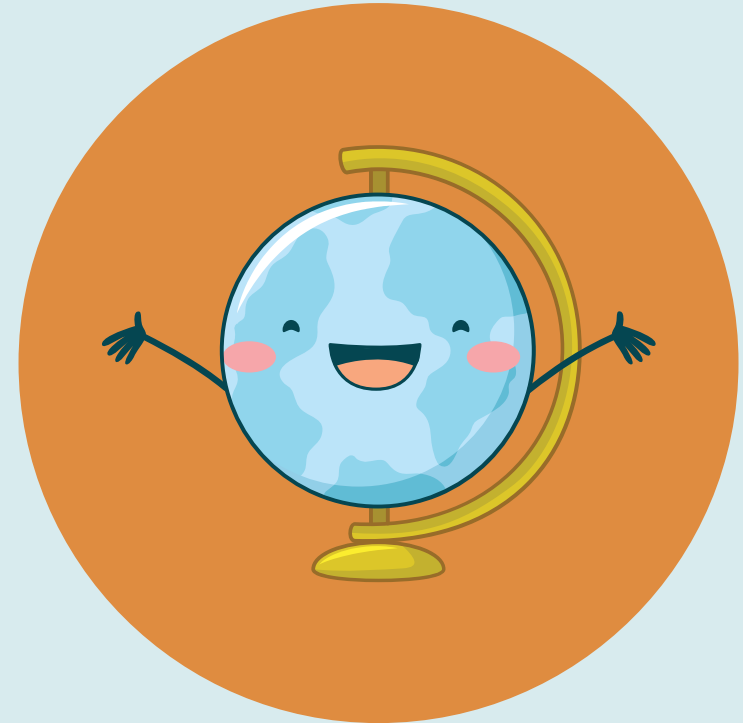
Book Metadata

Title: _____
Subtitle _____
Author: _____
Publisher: _____
Language: _____
Product Dimensions: _____
Shipping Weight: _____
Page count _____
ISBN-13: _____
Library of Congress # _____
Copyright # _____
Primary Subject _____
Keywords _____
Target Audience _____
Publication date _____
Price _____
Book Description: _____

Author Biography: _____

Source: <https://bookscouter.com/blog/book-metadata/>

Typical Digital Metadata Elements



Children’s Metadata Context

TABLE 2 Young children’s search elements and definitions

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D1	Animal	An animal	There is a gorilla.
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	Item	An item	It holds a key.
	Person	A person	A zookeeper
	Characteristic	The feature of a character or item	Its tail is very long.
	Shape	Shape	It is a rectangle.
	Amount	Amount	There are a lot of cookies.
	Color	Color	That chimpanzee is black.
	Place	A setting	In the zoo
	Chinese character	Chinese characters	I can read, so I know the Chinese characters for “cow” and “milk.”
	Layout	The specific layout of a page	Here is a big house.
	Size	Size	That wrong book is too big.
	Time	Time	When they say good night.
	Icon	Icon	A tiger icon.

(Source: Wang et al., 2022)

...More Context

Accessibility

Language	38	79.2%	132	9.0%		
Length	37	77.1%	108	7.4%		
Legibility	31	64.6%	51	3.5%		
Quantity	23	47.9%	40	2.7%		
Level of difficulty	13	27.1%	18	1.2%		
Text density	3	6.3%	3	0.2%	352	24.1%

Content

Illustrations	36	75.0%	105	7.2%		
Summary	21	43.8%	48	3.3%		
Topic	19	39.6%	47	3.2%		
Plot	24	50.0%	33	2.3%		
Genre	14	29.2%	26	1.8%		
Subject headings	4	8.3%	5	0.3%		
Style	4	8.3%	4	0.3%	267	18.3%

Engagement

Liking	17	35.4%	38	2.6%		
"Coolness"	18	37.5%	31	2.1%		
Mood	16	33.3%	28	1.9%		
Quality	14	29.2%	23	1.6%		
Interest	9	18.8%	12	0.8%		
Reading experience	8	16.7%	9	0.6%		
Visual appeal	5	10.4%	7	0.5%		
Interaction	2	4.2%	3	0.2%	152	10.4%

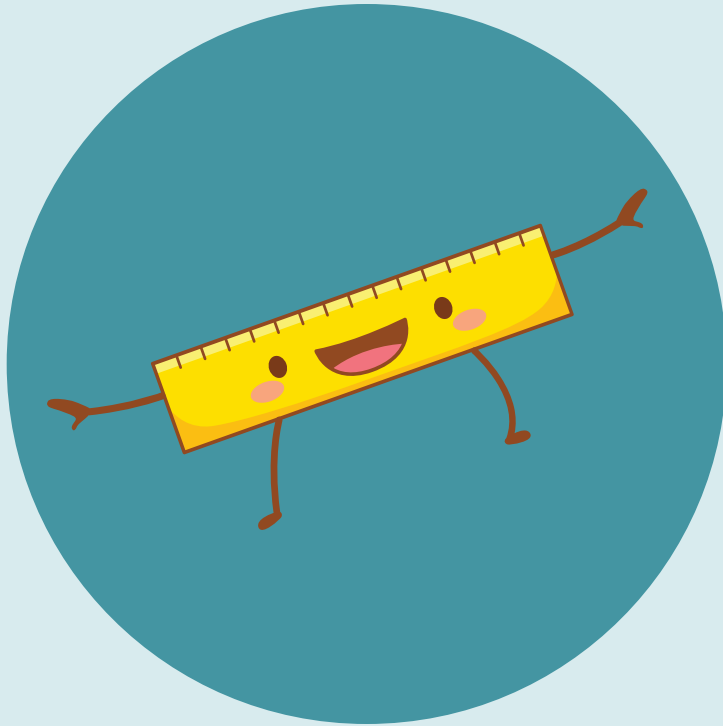
(Source: Reuter, 2007)

...and a bit more Context

Physical characteristics	• Book shape • Text arrangement • Texture on illustrations
Difficulty, Reading level, Understandability	• # of words or pages • Font size • Text size-density • Age appropriateness • Titles • Words inside books
Familiarity	• Prior experiences • Associated media • Recommendation • Language • Book shelves' location in the library
Emotional interest	• Internal factors of person • External factors of books
Personal characteristics	• Reading habits • Reading preferences • Reading motivation • Reading ability

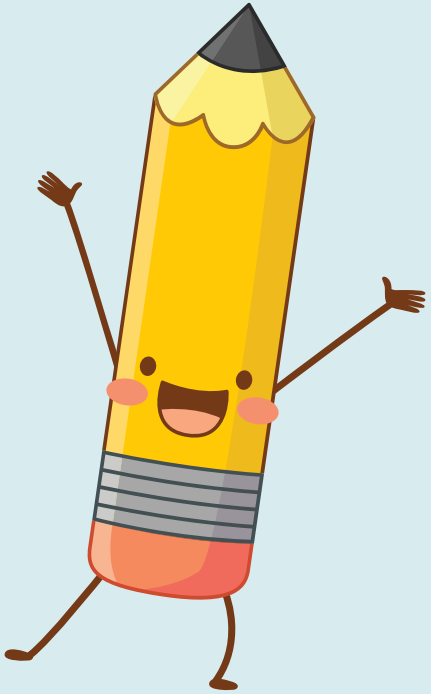
(Source: Beak, 2012)

Some Solutions



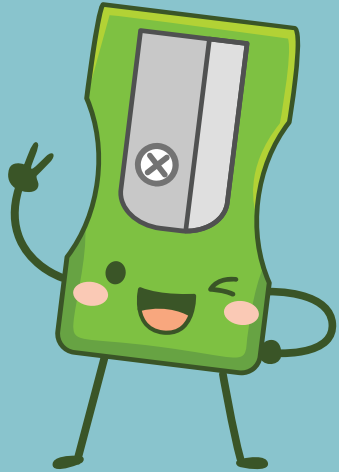
- ICDL - Proprietary metadata schema, digital library
 - Largely defunct
- Genre-fication, alternative/creative cataloging
 - BISAC – Topical subject headings arranged alphabetically
 - School librarians use BISAC/numeral hybrids
 - Metis - Designed to be developmentally appropriate & intuitive
 - “Orchestrated luck”
 - Picture labels, dots and stars sticker system, less hierarchy in 26 categories vs. DDC with 10

How Children Search, PK-4th



Often	Less Often
Browsing search	Catalog search
Aesthetic	Efferent
Simple classification	Complex classification
Intuitive	Academic

Questions?



- Thanks for listening!
- Please contact me at amanda.m.clark@du.edu if you'd like.
- To explore the information I found on this topic, [here is a link to my reference page!](#)